

Code-Layout, Readability and Reusability

Date	27 September 2026
Team ID	SWTID-2026-2825
Project Name	AI WeatherWise
Maximum Marks	5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	2-space indentation enforced across all .js files
2	Proper File Structure	Files and folders are logically organized	Yes	src/{config, controllers, middleware, models, routes, services}.plus app.js / server.js
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g. connectDB , userSchema , weatherService
4	Function / Method Names	Functions are descriptively named	Yes	e.g. registerUser , loginUser , getWeatherByCity , generateInsight
5	Code Comments	Inline and block comments explain logic	Partial	Core modules (db.js, controllers) are commented; small helper functions left self-explanatory
6	Modular Design	Code is split into reusable functions/modules	Yes	Routes / Controllers / Services / Models fully decoupled per the MVC pattern
7	No Redundant Code	Duplicate or unused code is removed	Yes	Shared logic centralized in weatherService.js / aiService.js
8	Error Handling	Exceptions and errors are handled gracefully	Yes	try/catch with centralized error handling and process.exit(1) on DB connection failure

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	authMiddleware.js	Node.js / Express	authRoutes, locationRoutes, weatherRoutes, aiRoutes - protects every private route	High
2	weatherService.js	Node.js (OpenWeatherMap client)	weatherController, aiController (weather context for the AI prompt)	High
3	aiService.js	Node.js (@google/genai)	aiController	High
4	db.js	Mongoose	server.js entry point	High
5	User.js / Location.js models	Mongoose Schema	authController, locationController, aiController	Medium

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear MVC separation under src/
Readability	4	Descriptive names throughout; comments could be denser inside services
Reusability	5	Middleware and services reused across all four route groups
Documentation / Comments	4	README.md plus inline comments in core modules (db.js, schemas)
Overall Score	4.5 / 5	Solid, maintainable, extensible codebase ready for the Testing phase